

e& Business Network Operations Guide

BNO Department Operations Manual

1. Introduction

The Business Network Operations (BNO) department at e& is responsible for managing and maintaining the telecommunications infrastructure that supports our network services. This document provides comprehensive guidelines for network operations, troubleshooting procedures, and best practices for ensuring optimal network performance and reliability.

2. Network Architecture Overview

The e& network infrastructure consists of multiple layers including core network, access network, and customer premises equipment. The core network operates on a redundant architecture with primary and secondary data centers located in Dubai and Abu Dhabi. Network traffic is routed through multiple paths to ensure high availability and fault tolerance.

2.1 Core Network Components

The core network includes routers, switches, firewalls, and load balancers. All core devices are configured with redundant power supplies and are connected to UPS systems. Network monitoring is performed 24/7 through our Network Operations Center (NOC).

2.2 Access Network

The access network provides connectivity to end users through various technologies including fiber-optic, DSL, and wireless connections. Fiber-optic connections are preferred for business customers due to higher bandwidth and reliability. DSL is used primarily for residential customers in areas where fiber deployment is not yet complete.

3. Network Monitoring and Management

Network monitoring is critical for maintaining service quality. The BNO team uses several monitoring tools including SNMP-based network management systems, flow analysis tools, and custom dashboards. Alerts are configured for various thresholds including bandwidth

utilization, latency, packet loss, and device availability.

3.1 Key Performance Indicators

The following KPIs are monitored continuously:

- Network Availability: Target 99.9% uptime
- Latency: Target < 50ms for local traffic, < 150ms for international
- Packet Loss: Target < 0.1%
- Bandwidth Utilization: Alert threshold at 80%
- Mean Time to Repair (MTTR): Target < 4 hours

4. Common Troubleshooting Procedures

4.1 Network Connectivity Issues

When investigating connectivity issues, follow these steps:

1. Verify physical connections and cable integrity
2. Check device status and interface statistics
3. Review recent configuration changes
4. Analyze network logs and error messages
5. Test connectivity using ping and traceroute
6. Check routing tables and ARP tables
7. Verify firewall rules and ACLs

4.2 Performance Degradation

Performance issues can be caused by bandwidth saturation, high CPU utilization on network devices, or routing problems. Use flow analysis tools to identify top talkers and bandwidth consumers. Review device CPU and memory utilization. Check for routing loops or suboptimal paths in the routing table.

5. Configuration Management

All network device configurations are stored in a centralized configuration management system. Before making changes, create a backup of the current configuration. All changes must be documented with a change request number and approved by the network operations manager. Configuration changes are tested in a lab environment before deployment to production.

5.1 Change Management Process

The change management process includes:

- Submit change request with justification
- Review and approval by technical lead
- Schedule maintenance window if required
- Backup current configuration
- Implement change during approved window
- Verify change and monitor for issues
- Update documentation

6. Network Security

Network security is paramount. All network devices must have strong passwords and SSH key authentication enabled. Access to network devices is restricted to authorized personnel only. Regular security audits are conducted to identify vulnerabilities. Firewall rules are reviewed quarterly to ensure they remain appropriate.

6.1 Access Control

Network device access is controlled through TACACS+ authentication. User accounts are created with appropriate privilege levels. All access attempts are logged and monitored. Privileged access requires additional approval and is granted on a temporary basis when possible.

7. Incident Response

When a network incident occurs, the BNO team follows a structured response procedure. Critical incidents are escalated immediately to the network operations manager. Incident tickets are created in the ticketing system with appropriate priority levels. Root cause analysis is performed for all major incidents to prevent recurrence.

7.1 Incident Severity Levels

Severity	Description	Response Time
Critical	Complete network outage affecting multiple customers	15 minutes
High	Partial outage or significant performance degradation	1 hour
Medium	Minor service impact or isolated issues	4 hours
Low	Non-critical issues or service requests	1 business day

8. Maintenance Windows

Scheduled maintenance is performed during approved maintenance windows. Standard maintenance windows are:

- Weekdays: 2:00 AM - 4:00 AM (UAE time)
- Weekends: 12:00 AM - 6:00 AM (UAE time)
- Emergency maintenance: As needed with customer notification All maintenance activities must be coordinated with the NOC and customer support teams. Customers are notified at least 48 hours in advance for planned maintenance.

9. Contact Information

For network operations support, contact:

- NOC Hotline: +971-XX-XXX-XXXX
- Email: bno-noc@eand.ae
- Emergency: +971-XX-XXX-XXXX (24/7)
- Network Operations Manager: Available during business hours For configuration changes and access requests, submit tickets through the IT service management portal.

Appendix A: Common Network Commands

Useful commands for network troubleshooting:

Ping: ping -c 4 destination_ip

Traceroute: traceroute destination_ip

Show Interface: show interface gigabitEthernet 0/0

Show IP Route: show ip route

Show ARP: show arp

Show Running Config: show running-config

Show Logs: show log | tail -100

Appendix B: Network Device IP Ranges

Standard IP addressing scheme:

- Core Routers: 10.0.0.0/24
- Access Switches: 10.0.1.0/24
- Firewalls: 10.0.2.0/24
- Load Balancers: 10.0.3.0/24
- Management Network: 172.16.0.0/24 All devices use /24 subnets for management interfaces. Customer-facing interfaces use public IP addresses as assigned by RIPE NCC.